Rafat Abdullah

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Objective

Computer Science undergraduate seeking an internship at Huawei Technologies Bangladesh to apply hands-on experience in software development, applied AI, and systems-level programming to real-world engineering problems, while growing within a large-scale technology organization.

Education

Islamic University of Technology (IUT), Gazipur, Bangladesh — Expected Graduation: 2027
B.Sc. in Computer Science and Engineering (CSE) — Currently in 3rd Year, 2nd Semester
CGPA: 2.75 / 4.00

Technical Skills

Programming Languages: Python, C, C++
Database Compatibility: MySQL, Oracle, MongoDB, PostgreSQL

Projects

VisionCart — Visual Inventory Management System — GitHub
Python, FastAPI, SQLite/SQLAlchemy, Google GenAI SDK (Gemma vision-language model)

- Built a full-stack system that replaces manual shelf counting: a photo of a shelf is analyzed and reconciled against a live inventory database in real time.
- Integrated a vision-language model (Gemma) via Google's GenAI SDK to detect products and count units directly from images, automatically flagging low-stock and out-of-stock items.
- Designed a FastAPI backend with a SQLAlchemy-backed database and a responsive vanilla JS/HTML/CSS dashboard (live KPIs, search/filter/sort, light & dark themes); deployed live on Render.

My Pocket Buddy — On-Device AI Companion App — GitHub
TypeScript, React Native, Expo, llama.rn (on-device LLM inference), MMKV

- Built a cross-platform (iOS/Android) mobile app that runs a local LLM entirely on-device via `llama.rn` (GGUF models, CPU-only inference) for private, offline AI chat with no cloud API dependency.
- Designed a hybrid context-injection system for the chat engine: a sliding window of recent turns plus regex-triggered full-history recall, so the AI companion can reference earlier conversations.
- Implemented a voice-activated alarm that listens via microphone to detect when the user wakes up and dismiss it automatically, alongside a stats dashboard and an AI-generated exam-practice screen.

PMVC-WNM — Banglish Sentiment Classification (Co-training Prototype) — GitHub
Python, scikit-learn — Course project, CSE 4622 Machine Learning Lab, IUT

- Implemented a two-view co-training pipeline (character n-gram SVM + phonetic-encoding logistic regression) to classify sentiment in Banglish (Romanized Bangla-English code-mixed) text as a lightweight alternative to deep learning.
- Designed a rule-based phonetic encoder to normalize inconsistent Romanized Bangla spellings (e.g. *khacchi* / *khacci* / *khachi*) into shared phonetic codes.
- Added a weakly supervised noise model that down-weights unreliable pseudo-labels using cross-view disagreement, aiming for competitive accuracy with no GPU and orders of magnitude fewer parameters than a deep-learning baseline.

Local Judge for Windows — Offline Competitive Programming Judge — GitHub
Python, C++, JavaScript

- Built a local judge that compiles and runs Python/C++ submissions against test cases under configurable time and memory limits, without depending on any Linux-only tooling.
- Implemented real-time process time/memory tracking and verdict reporting (Accepted, Wrong Answer, Runtime Error, Time Limit Exceeded), mirroring the behavior of an online judge for local, offline practice on Windows.

Achievements

- **Competitive Programming:** Codeforces rating 885 (handle: NebulaKnight); selected for the NHSPC national round twice
- **Hackathons:** IUT-RS Hackathon; Build with Gemma — ML, AI, Deep Learning & NLP Community (Kaggle)